

**YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCE
TOPIC TEST**

Name: SOLUTIONSMark: 53**Part 1: Multiple Choice**

Write your choice for each question in the table below.

1. The crumple zone of a car is designed to:
 - (a) allow the vehicle and passengers to stop more slowly.
 - (b) decrease the time before the car comes to a stop after a collision.
 - (c) allow the chassis to bend.
 - (d) create a reaction force that equals the action force.
2. How long would a car travelling at a speed of 20 m/s take to cover a distance of 360 m?
 - (a) 13 s
 - (b) 26 s
 - (c) 18 s
 - (d) 34 s
3. A car manufacturer wants to produce cars that accelerate faster. Which changes in their cars would result in greater acceleration?
 - (a) Increase both the forward force by the engine and the mass of the car.
 - (b) Decrease both the forward force and the mass of the car.
 - (c) Increase the forward force and decrease the mass of the car.
 - (d) Decrease the forward force and increase the mass of the car.
4. Speed is a measure of the rate:
 - (a) at which distance is covered.
 - (b) at which speed increases.
 - (c) of change in force.
 - (d) of change in acceleration.
5. The rate of change of speed of an object is known as:
 - (a) average speed.
 - (b) average velocity.
 - (c) change in velocity.
 - (d) acceleration.
6. An object starts from rest and accelerates at 5.0 m/s^2 . After 4.0 seconds, its speed will be:
 - (a) 0
 - (b) 10 m/s.
 - (c) 20 m/s.
 - (d) 30 m/s.

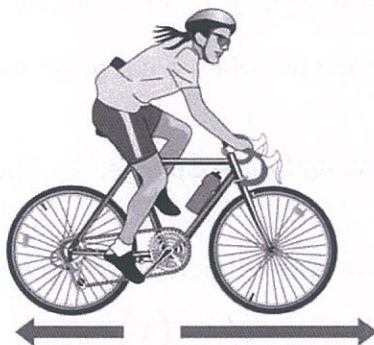
MULTIPLE CHOICE ANSWERS	
1	A
2	C
3	C
4	A
5	D
6	C
7	A
8	A
9	A
10	B
11	C
12	B
13	B
14	B
15	A
16	C
17	A
18	A

7. The tendency of an object to resist changes in its motion is called:
- (a) inertia.
 - (b) weight.
 - (c) friction.
 - (d) work.
8. Newton's law, known as the law of inertia, is:
- (a) his first law.
 - (b) his second law.
 - (c) his third law.
 - (d) none of the above.
9. Belinda walks a total distance of 1200 metres to school each morning. If this trip takes her 800 seconds, her average speed would be:
- (a) 1.5 m/s.
 - (b) 0.67 m/s.
 - (c) 0 m/s.
 - (d) 960,000 m/s.
10. An object is acted upon by a force of 50 N pushing it forwards and a total frictional force of 40 N acting in the opposite direction. The nett force will be:
- (a) 10 N backwards.
 - (b) 10 N forwards.
 - (c) 90 N backwards.
 - (d) 90 N forwards.
11. The size of the acceleration acting on an object of mass 10 kg that has a nett force of 3.0 N acting on it will be:
- (a) 13 m/s^2 .
 - (b) 3.33 m/s^2 .
 - (c) 0.3 m/s^2 .
 - (d) 30 m/s^2 .
12. A car is driven along a highway at an average speed of 90 km/h for 2 hours without a rest stop. What distance has it covered?
- (a) 45 km
 - (b) 180 km
 - (c) 360 km
 - (d) Not enough information is given to calculate.
13. The force of gravity acting on an object is called:
- (a) inertia.
 - (b) weight.
 - (c) friction.
 - (d) work.

14 The weight force acting on a 20 kg mass on the Earth would be:

- (a) 19.8 N
- (b) 198 N
- (c) 1.98 N
- (d) 0 N.

15. In the diagram below, the arrows show the size and direction of two force arrows acting on the bike.



Which of the following describes the effect of the pair of forces on the bike?

- (a) The bike will accelerate forward.
 - (b) The bike will change its direction of motion.
 - (c) The bike will remain stationary.
 - (d) The bike will decelerate.
16. What is the acceleration of a car if it increases its velocity from 5.0 m/s to 15.0 m/s in 4.0 seconds?
- (a) 2.5 m/s
 - (b) 15 m/s²
 - (c) 2.5 m/s²
 - (d) 5.2 m/s²
17. A class is studying Newton's three laws. The teacher asks a student wearing rollerblades to face the wall and push against it. The student moves backwards away from the wall. Which is the best explanation for her movement?
- (a) Her action of pushing resulted in a reaction force by the wall, in the opposite direction to her push.
 - (b) She used her feet to push herself away from the wall.
 - (c) The push is a force and causes her to accelerate.
 - (d) The inertia of the student is responsible for her acceleration.
18. Which of the following statements is **correct**?
- (a) A car at rest has at least two forces acting on it.
 - (b) A car at rest has no forces acting on it.
 - (c) When you take your foot off the car's accelerator, the car slows down because there are no more forces acting on it.
 - (d) When you brake in a car that is rolling down a hill, the car stops because all of the forces add up to zero.

Part 2: Short Answer

1. Decide whether each of the following statements is **true (T)** or **false (F)**. (4 marks)

(a) Average speed is calculated by distance covered multiplied by the time taken.

F

(b) $40.0 \text{ km/h} = 144 \text{ m/s}$

F

(c) When a car accelerates quickly, you move forward in your seat.

F

(d) A negative acceleration occurs when a car goes backwards.

F

[1 mark each]

2. The map below shows Broome and Alice Springs. Use the map and its scale to work out the following.

(a) The **distance** between Broome Airport and Alice Springs airport.

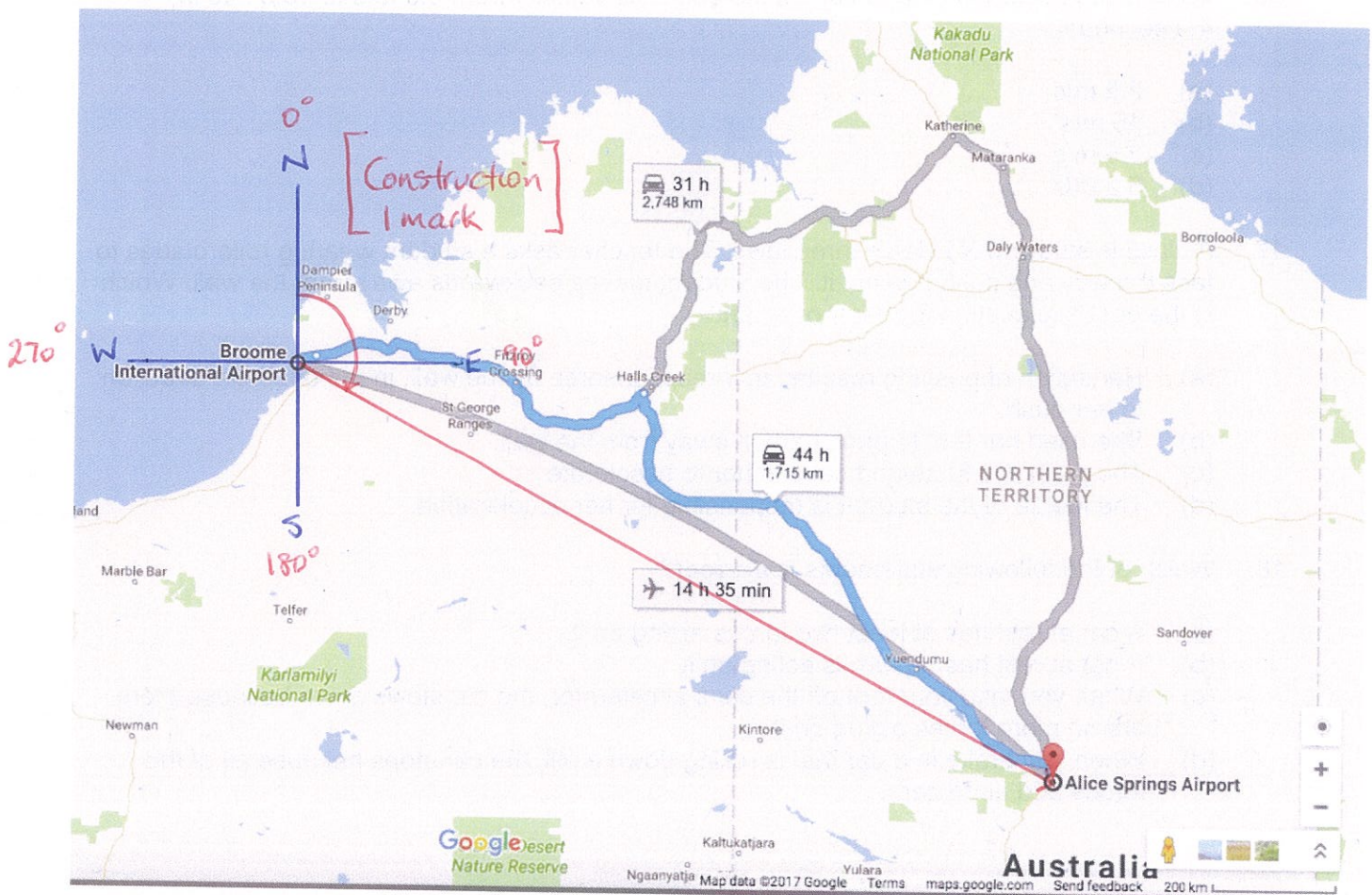
(1 mark)

1715 km (1)

(b) The **displacement** between the two airports. (HINT: Don't forget to include a **bearing**.)

(3 marks)

1435 km ($\pm 100 \text{ km}$) bearing 120° ($\pm 10^\circ$)
(1) (1)



3. Complete the following table on **speed** calculations. Make sure you include the appropriate units for each quantity.

(3 marks)

Distance travelled	Time taken	Average speed
490 m	35 s	14 m/s
276 m	12 s	23 m/s
121 m	11 s	11 m/s

[1 mark each]

4. Explain why it is dangerous to have large parcels or sharp objects loose in the back of a moving car. (2 marks)

- Object has inertia and continues to move forward. (1)
- Can seriously injure passengers if they are hit. (1)

5. Describe how a front air bag protects the driver of a car that has stopped suddenly as a result of a collision. (2 marks)

- The airbag collapses as the body hits it. (1)
- This slows the body down and decreases the forces experienced. (1)

[Could also mention that it prevents the person hitting the steering wheel or dashboard.]

6. Describe the motion of a ball placed on the middle of the back seat of a car in the following situations.

- (a) The car brakes suddenly. Moves forwards. (1 mark)
- (b) The car turns right. Moves left. (1 mark)
- (c) The car accelerates. Moves backwards (1 mark)

[1 mark each]

7. A dragster accelerates from rest at 14.5 m/s^2 for 6.90 s during a race. What is the velocity of the vehicle after this time? (**Set your working out clearly.**) (3 marks)

$$v = ?$$

$$u = 0 \text{ m/s}$$

$$a = 14.5 \text{ m/s}^2$$

$$t = 6.90 \text{ s}$$

Forwards is +ve.

$$v = u + at \quad (1)$$

$$= 0 + (14.5)(6.90) \quad (1)$$

$$= \underline{100 \text{ m/s forwards}} \quad (1)$$

8. Complete the following table on **force** and **acceleration**.

(3 marks)

Force	Mass	Acceleration
6000 N	1200 kg	5 m/s ²
150 N	25 kg	6 m/s ²
60 N	24 kg	2.5 m/s ²

[1 mark each]

9. The Space Shuttle lands at 270 km/h (75 m/s) and immediately pops a drag chute to assist its brakes to stop the vehicle. Assume that the braking forces are uniform for this motion.



- (a) If it takes 15 s to stop, calculate the **deceleration** experienced by the crew. (3 marks)

$$v = 0 \text{ m/s}$$

$$u = 75.0 \text{ m/s}$$

$$a = ?$$

$$t = 15 \text{ s}$$

Forwards is +ve

$$a = \frac{v - u}{t} \quad (1)$$

$$= \frac{0 - 75.0}{15} \quad (1)$$

$$= -5.00 \text{ m/s}^2$$

$$\therefore \text{Deceleration} = 5.00 \text{ m/s}^2 \text{ backwards.} \quad (1)$$

- (b) If the mass of the Shuttle is 34,500 kg, what is the average **force** exerted by the brakes and the chute?

(If you couldn't find an answer for part (a), use deceleration = - 6.0 m/s².)

(3 marks)

$$F = ?$$

$$m = 34,500$$

$$a = -5.00 \text{ m/s}^2$$

Forwards is +ve

$$F = ma \quad (1)$$

$$= (34,500)(-5.00) \quad (1)$$

$$= -172,500 \text{ N}$$

$$\therefore F = 172,500 \text{ N backwards.} \quad (1)$$

10. An object has a mass of 6.0 kg.

(a) What is its **weight** on Earth?

(2 marks)

$$F_w = ?$$

$$m = 6.0 \text{ kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$F_w = mg$$

$$= 6.0 \times 9.8 \quad (1)$$

$$= \underline{58.8 \text{ N}} \quad (1)$$

(b) Would the object **weigh more or less** on the Moon? Explain your answer.

(2 marks)

• Less. (1)

• Less gravity so "g" is less. (1)

(c) What is its **mass** on the Moon?

(1 mark)

$$6.0 \text{ kg} \quad (1)$$